Fiscal Year:	FY 2006	**Task Last Updated:**	FY 09/20/2006
PI Name:	Dryer, Frederick L Ph.D.		
Project Title:	Scientific Support for a Space Shuttle Droplet Burning Experiment		
Division Name:	Physical Sciences		
Program/Discipline:	COMBUSTION SCIENCE		
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Droplets, sprays, particles, dust clouds, nanoparticles		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	fldryer@princeton.edu	**Fax:**	FY 609-258-6109
PI Organization Type:	UNIVERSITY	**Phone:**	609-258-5206
Organization Name:	Princeton University		
PI Address 1:	Mechanical & Aerospace Engineering		
PI Address 2:	D-329-D Engineering Quad		
PI Web Page:			
City:	Princeton	**State:**	NJ
Zip Code:	05844-5263	**Congressional District:**	12
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	OTHER
Start Date:	08/01/2004	**End Date:**	07/31/2006
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
Contact Monitor:	Nayagam, Vedha	**Contact Phone:**	
Contact Email:	Vedha.Nayagam@lerc.nasa.gov		
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNC04AA66A		
Performance Goal No.:			
Performance Goal Text:			
Task Description:	This program supports the engineering, test matrix refinement, data analysis, and data interpretation requirements for the study of initially single component, spherically symmetric, isolated droplet combustion studies through experiments and numerical modeling. The work is presently being expanded to also support similar development and analyses of experiments conducted with slow axi-symmetric convection present. Efforts are coordinated with the contributions of Prof. F. A. Williams, Prof. , University of California, San Diego (PI), Prof Mun Choi. of Drexel University, and Prof. Benjamin Shaw of University of California, Davis, and with NASA personnel, Mr. Michael Hicks, NASA Glenn Research Center and Dr. Vedha Nayagam, National Center for Space Exploration Research, as part of a new flight program, Flame Extinguishment Experiment (FLEX) described in Williams et al. (2005). The purposes of this program are multi faceted, having both fundamental and applied impacts. The principal objective of the work is to develop specific information that assists in defining appropriate fire safety approaches and procedures for human exploration endeavors. The work simultaneously provides scientific fundamental knowledge on droplet combustion dynamics for		

pure fuels and fuel-water mixtures that impact efficient use of transportation fuels and chemical propellants in space with minimum environmental impact. The contributions to the overall program by this project in the last year have been to make first assessments of the effects of diluents on the extinction behavior of burning methanol and n-heptane droplets using earlier models, and to extensively modify numerically modeling aspects to include: effects of internal liquid phase motion on droplet burning phenomena in a more rigourous manner; improve numerical methods so that larger detailed kinetic mechanisms can be utilized in calculations; add sooting model components to the overall numerical model construct. As the reporting period ended, the numerical methodology updates have been completed, a pseudo-unidimensional embodiment of multi-dimensional internal liquid motion effects has been developed and is ready for implementation; and initial sooting mechanisms have been investigated but are not fully implemented.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

Although this experimental/theoretical investigation is developed specifically to yield fire safety engineering guidance, the experimental and theoretical information gathered has direct impact on combustion properties on earth associated with the efficient use of ground transportation, air transportation, and chemical rocket propulsion systems.

Introduction
The combustion characteristics of isolated liquid fuel droplets of n-heptane, n-decane, methanol, methanol-water, ethanol and ethanol-water having initial diameters between about 1 mm and 6 mm are investigated in both 1-dimensional, non-convective, and two-dimensional, slow convective gaseous oxidizer composed of oxygen and various inert diluents, including nitrogen, helium and carbon dioxide are used to evaluate the relative and absolute effectives of different atmospheric diluents that are candidates for fire suppression in low gravity environments. Experiments conducted on the International Space Station and in ground based low gravity facilities will provide data that through numerical and analytical interpretation will yield results that define the effectiveness of various fire control approaches to be applied in exploration environments.

Because of reduced or entire lack of buoyancy forces in exploration environments, the propagation and extinguishment characteristics of fire are different than those found in earth's gravity field. As a result, the effectiveness of various methods for controlling fire spread and extinguishment need to be better defined through experimental testing and theoretical analyses which clearly define these changes. Isolated droplet burning environments, while not used in fire research approaches in earth's gravity provide an excellent mechanism for assessing important parameters with minimum resources, experimental simplicity, their ability to be numerically and analytically modeled.

Although this experimental/theoretical investigation is developed specifically to yield fire safety engineering guidance, the experimental and theoretical information gathered has direct impact on combustion properties on earth associated with the efficient use of ground transportation, air transportation, and chemical rocket propulsion systems.

Progress

The complexity of the numerical modeling task has substantially increased in order to include autoignition behavior, sooting characteristics, and slow convection components for analysis of droplet burning. While the existing numerical codes we have developed can accommodate simple sphero-symmetric problems, these additional complexities have required major revisions of the prior code to address facets associated with the FLEX program. These revisions include updating all sub-model aspects of the simulation, as well as the numerical and data storage aspects of the prior approach. At present, the code revisions have been validated against single-component droplet combustion calculations using the prior code with a regressing liquid surface, detailed transport and chemistry in the gas phase and thermal transport in the liquid phase but without the inclusion of radiative effects. Gas-phase transport phenomena are simulated using the publicly-available EGLIB software package (Ern and Giovangigli, 1996). Integration of non-luminous radiative transport, arbitrary liquid-phase composition, adaptive (r-type) grid modification for global error minimization and soot growth, transport and oxidation are presently being completed.

Coupled with these revisions are efforts to understand and characterize how longer time scale liquid motions, on the order of tenths of seconds to seconds, affect burning histories and extinction for large droplets. Understanding how these multi-dimensional phenomena can be integrated without yielding inordinate computational resources is a significant challenge, especially as one wishes to continue to numerically simulate the gas phase chemistry, radiative coupling and transport phenomena accurately. We have been developing an approach to integrate multi-dimensional liquid phase transport phenomena into an essentially uni-dimensional gas phase numerical description coupled at the liquid-gas interface. Internal circulation has a significant and well-documented effect on evaporation and combustion of liquid droplets. Internal circulatory motions in small (~20 microns diameter) droplets are quickly damped by viscosity, but larger (~1 mm diameter) droplets tend to be of greater research interest due to longer droplet lifetimes and greater droplet inertia. Study of droplet combustion in low gravity environments is required in order to obtain reasonable temporal and spatial resolution of experimental observations, and to observe radiative extinction. These motions can be imparted by generation techniques as well as arise from fluid instabilities resulting from surface tension gradients with temperature and component concentration. We have developed a two-dimensional computational model which fully resolves the flow field for both isolated and mechanically supported droplets has been developed. The approach will allow for proper interpretation of small droplet phenomena via large droplet experiments, and computationally, the approach permits continued detailed description of the gas phase phenomena that are important to defining fire safety related combustion behavior.

The model (Kroenlein et al., 2005) includes variable thermo-physical properties and free surface boundary conditions consistent with a combustion environment. The data from this model is then analyzed to calibrate both a quasi-one-dimensional liquid mixing model for a free droplet in quiescent ambient with no body force. Viscous, thermo-capillary and diffuso-capillary effects are treated as significant. Surface tension is assumed to be linear with component concentration. An in-house two-dimensional, axisymmetric hydrodynamic code which accurately resolves liquid phase behaviors and calculates gas-phase behaviors utilizing a semi-empirical model calibrated with the detailed uni-dimensional droplet combustion code described above.

With the above revisions completed, numerical simulations can be compared with a number of microgravity experiments already conducted on the burning of methanol and hydrocarbon droplets in air diluted with various inerts as well as in oxygen-inert mixtures in which another diluent is substituted for nitrogen and the oxygen index is held

Task Progress:	constant. The FLEX program will utilize methanol and n-heptane as test fuels. We have further improved the detailed kinetics for these two fuels over the past year. Under another program in this laboratory, we have further updated small molecule and radical chemistry, including that for methanol (Li et al, 2006), and we have also advanced our work on the chemical kinetics of n-heptane. As part of earlier work which is being readied for inclusion in Mr. Brad Urban's PhD thesis (to be submitted later this year), we have developed new methodology to investigate what controls the thermal autoignition of n-heptane/air mixtures (Kazakov et al, 2006). Computational Singular Perturbation (CSP) analysis can be used to gain understanding of the complex kinetic behavior associated with two-stage ignition of large hydrocarbon molecules. To this end, available detailed and reduced chemical kinetics models commonly used in numerical simulations of n-heptane oxidation phenomena were directly analyzed to interpret the underlying fundamental steps leading to two-stage ignition. Unlike previous implementations of the CSP methodology, temperature was included as one of the state variables so that factors controlling ignition can be unambiguously determined. The analyzed models showed differences in the factors contributing to the initial development and shutdown of the first ignition stage. However, during the second stage, both models show the importance of the degenerate branching decomposition of hydrogen peroxide, which contradicts some previous interpretations of this phenomenon. Further analyses show that depending on how one depicts the detailed chemical kinetics, the species distribution at the end of the first stage of ignition can be significantly different than that produced by a detailed kinetic model. This CSP analysis method is currently being applied to understand in more detail what controls autoignition of isolated droplets of n-heptane in air under microgravity conditions. This work will appear in the PhD thesis of Mr. Brad Urban (in preparation). We have developed a new minimized model for n-heptane, iso-octane, and their mixtures (Chaos et al., 2006) for use in high temperature applications such as premixed and diffusion flame environments. This model will be used in future computations for n-heptane droplet burning and extinction studies. We have also continued to refine the kinetic mechanisms that can be used in numerical droplet combustion calculations with ethanol (Li et al, 2005), n-decane (Zhao et al, 2005), or n-hexadecane (Chaos, et al., 2005) as fuels. In fact, we have produced a generic modeling approach that can be used to simulate the high temperature chemical kinetics of pure n-alkanes with carbon numbers from 7 to 16, as well as their mixtures. These new mechanisms and the above code can be utilized to model other existing data on isolated droplet burning developed by Dietrich and co-workers as well as by Shaw and co-worker. Finally, in work associated with the present transition to the FLEX program, we collaborated with Dr. M.Y Choi and colleagues and Dr. E. Ranzi and colleagues to study time dependent, spherically symmetric combustion of an isolated ethanol droplet under microgravity conditions that included sooting effects within the computation (Mehl et al., 2005). Sooting was integrated into the chemical kinetic model using a sectional modeling approach., and the particulate radiation was incorporated via a contribution to the local Plank mean absorption coefficient. Results compared favorably with recent experimental data. Burning rates and extinction diameters were correctly predicted with varyiations in the initial droplet diameter. The model captured the major burning features (such as burning rate and the extinction diameter) and to reasonably characterize the formation of PAH and soot. The relative amounts of soot and its precursors were in line with the experimental observations in terms of their distance from the droplet surface and the flame front. Similar to the experimental evidence with which the computations were compared in the paper, oxygen index exhibited a non-monotonic effect on peak soot volume fraction. Although these results are very encouraging in terms of including sooting in the droplet computational predictions, quantitative comparisons remain to be developed. The peak soot volume fraction observed experimentally was under-predicted by an order of magnitude. These deficiencies can be partially attributed to the significant uncertainties remaining in the physical submodels, including that for ethanol kinetics, the PAH formation mechanism and the soot model itself. Despite these uncertainties, the work is promising in terms of including soot formation effects in the analyses of FLEX experimental results and for producing improved sub-models for fire safety applications. References Chaos, M. Kazakov, A., Zhao, Z., Zeppieri, S.P., and Dryer, F.L. (2005). "Reduced High-Temperature Mechanisms for Large Paraffins – n-Hexadecane", Eastern States Section Fall Meeting, the Combustion Institute, Orlando, FL November 14,15. Extended Abstract Chaos, M., Kazakov, A., Zhao, Z., and Dryer, F.L. (2006). "A High Temperature Chemical-Kinetic Model for Primary Reference Fuels", Int J. Chem. Kin, Submitted Dietrich, D.L., Struk PM, Ikegami, M, Xu (2005). "Single Droplet Combustion of Decane in Microgravity: Experiments and Numerical Modelling Combustion Theory and Modelling, 4: 569-585. Ern, A. and Giovangigli, V. (1996). "Optimized Transport Algorithms for Flame Codes", Combust. Sci. and Tech. 118, 387-396. Kazakov, A., Chaos, M., Zhao, Z., and Dryer, F.L. 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